DBMS Assignment 1

ToyDB: PF Buffering, Slotted Pages, and B+ Tree Indexing

Implementation and Experimental Report

1. Overview

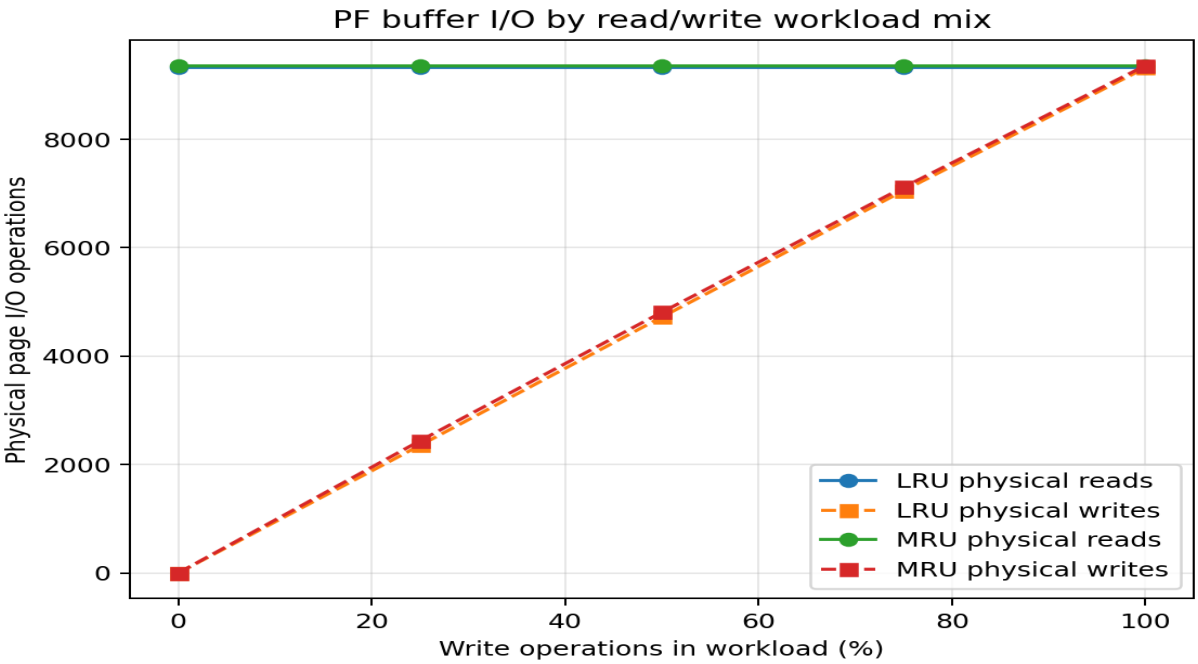
This project implements configurable LRU/MRU page buffering in the PF layer, a variable-length slotted-page heap file in the HF layer, and Student roll-number indexing in the AM layer. All experiments use 17,813 non-empty records from the supplied Student dataset.

2. PF page buffering

The PF buffer pool supports a configurable frame count, LRU and MRU replacement, dirty-page writeback, and counters for logical and physical I/O, hits, misses, and evictions. The benchmark uses 128 pages, an 8-frame pool, 10,000 operations per workload, and write shares from 0% to 100%.

Policy	Write %	Logical R	Logical W	Physical R	Physical W	Hits
LRU	0	10,000	0	9,319	0	681
LRU	25	10,000	2,500	9,319	2,354	681
LRU	50	10,000	5,000	9,319	4,721	681
LRU	75	10,000	7,500	9,319	7,053	681
LRU	100	10,000	10,000	9,319	9,319	681
MRU	0	10,000	0	9,360	0	640
MRU	25	10,000	2,500	9,360	2,439	640
MRU	50	10,000	5,000	9,360	4,810	640
MRU	75	10,000	7,500	9,360	7,116	640
MRU	100	10,000	10,000	9,360	9,360	640

For this deterministic random-access workload, LRU produced 41 fewer physical reads than MRU (9,319 versus 9,360). The difference is small and is specific to this access pattern.

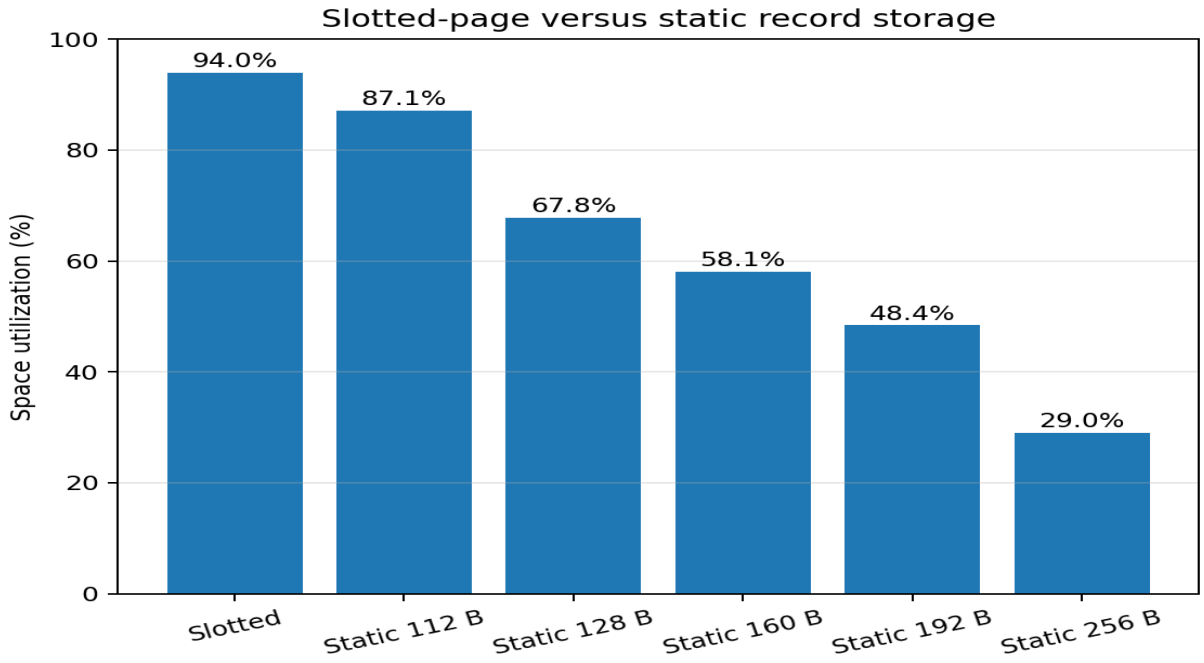

PF buffer I/O by read/write workload mix
Physical page I/O operations
0
2000
4000
6000
8000
0
20
40
60
80
100
Write operations in workload (%)
LRU physical reads
LRU physical writes
MRU physical reads
MRU physical writes

3. HF slotted-page storage

The HF layer stores variable-length text records in pages with a header and slot directory. Deletion invalidates a slot; compaction before a subsequent insertion reclaims fragmented space while preserving live record RIDs. The benchmark inserted and sequentially scanned all 17,813 records.

Method	Max record length	Pages	Allocated bytes	Utilization
Slotted	Variable	1,921	1,959,420	94.00%
Static	112 B	1,980	2,019,600	87.10%
Static	128 B	2,545	2,595,900	67.76%
Static	160 B	2,969	3,028,380	58.08%
Static	192 B	3,563	3,634,260	48.40%
Static	256 B	5,938	6,056,760	29.04%

The slotted-page representation uses fewer pages than every tested static record length and reaches approximately 94% utilization. Static allocation becomes less efficient as the maximum record length grows.



4. AM B+ tree indexing

The second semicolon-separated Student field is an 8-character roll number and is indexed as a fixed-width character key. Duplicate roll numbers are retained using RID overflow chains. The input row number is used as the record identifier.

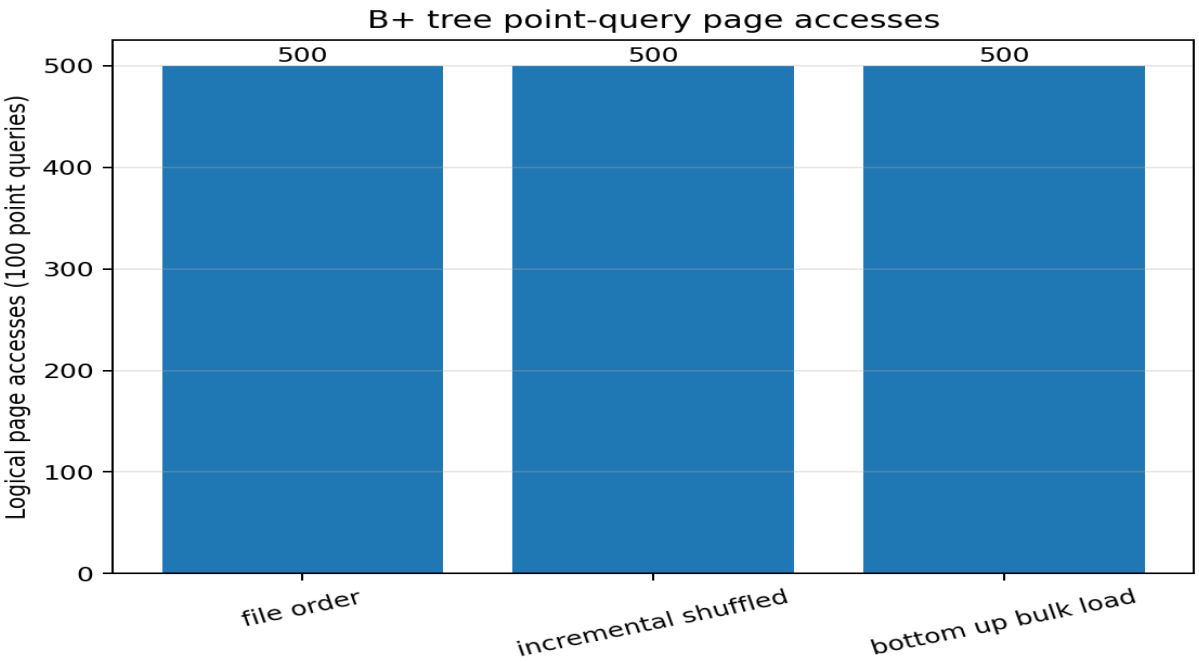
Construction method	Build time (s)	Logical page accesses	Physical reads	Physical writes
Existing-file pass	0.021132	50,492	289	748
Incremental shuffled	0.053679	50,172	15,156	15,085
Bottom-up bulk load	0.001056	289	289	583

The bottom-up loader sorts keys, packs leaf pages including duplicate RID chains, and constructs internal levels from child minimum keys. This avoids repeated tree searches and split operations during construction.

Method	100-query time (s)	Logical page accesses	Physical reads	Correct
Existing-file pass	0.003398	500	98	100/100
Incremental shuffled	0.003757	500	115	100/100
Bottom-up bulk load	0.003359	500	41	100/100

All three methods returned the correct record for 100 deterministic equality lookups. The bottom-up index also passed a full sequential scan returning all 17,813 record IDs.




B+ tree point-query page accesses
Logical page accesses (100 point queries)
0
100
200
300
400
500
500
500
500
file order
incremental shuffled
bottom up bulk load

5. Validation and reproduction

Run **make clean && make** from the repository root to compile. Run **./run_all_experiments.sh** to reproduce the PF, HF, and AM measurements and regenerate the CSV files and figures. The HF test program checks insertion, retrieval, deletion, deleted-slot rejection, slot reuse after compaction, and sequential scanning.

6. Measurement notes

CPU timings are single-run measurements and vary by machine and filesystem. The PF benchmark uses a deterministic pseudo-random access sequence; its LRU/MRU ranking should not be generalized to all workloads. The bottom-up bulk loader is specialized to the assignment dataset and fixed-width 8-byte character key.